



CLIMATE CHANGE IMPACT

ON ANDAMAN AND NICOBAR ISLANDS

Ocean and Global Change

Jaidev Sanjay Khalane

Preface

In the vast expanse of the Indian Ocean lies a jewel of biodiversity and a treasure of scenic views - the Andaman and Nicobar Islands. As the world grapples with the repercussions of climate change, these islands stand as a microcosm of the challenges and opportunities that lie ahead. Our project delves into the analysis of the intricate web of interactions between rising sea levels, changing weather patterns, the delicate ecosystems of these islands, and many more factors and their implications.

Through a multidisciplinary approach, we also aim to unravel the mysteries of how climate change is reshaping the very fabric of life on these islands. From analysing the factors influencing the Andamans to the impact of them on these islands. We will also discuss and analyse the results of the simulation of the impact of these events on the landscape of the Andaman Islands.

Acknowledgements

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Thank You

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1 Introduction

The Andaman and Nicobar Island group is an important union territory of India situated in the Bay of Bengal and the Andaman Sea region. It consists of a group of 836 Islands in two smaller sections named the Andaman Islands and the Nicobar Islands. The Andaman and Nicobar Islands, respectively are the northern and southern parts of the island cluster, and they are separated by a 150 km long channel called the 10-degree channel. The islands extend from 6 degrees to 14 degrees North latitudes and from 92 degrees to 94 degrees East longitudes. The Andaman Islands cover an area of 6,048 kilometres square while the Nicobar Islands cover an area of 1,841 kilometres square. The highest point at present is the Saddle Point, at an elevation of 737 m. The only active volcano in India, that is, the Barren Island is present in the Andaman Sea and belongs to the Andaman Islands. The islands have a coastline of length 1,962 km.

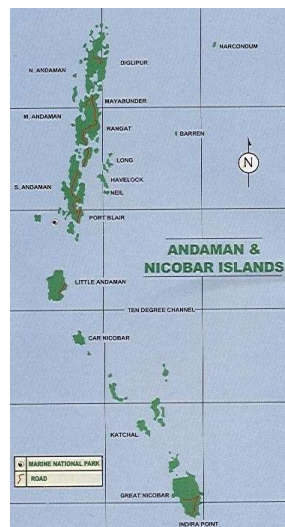


Fig. 1. Map of Andaman and Nicobar Islands. Source [15]

The topography of the territory includes a large diversity of landscapes ranging from sandy to rocky sandstone to beaches and coral reefs. There is also a significant variation

in the altitude of the island starting with almost completely flat Islands and gradually raising topography to higher interior lands. This can be clearly seen from the figure given below.

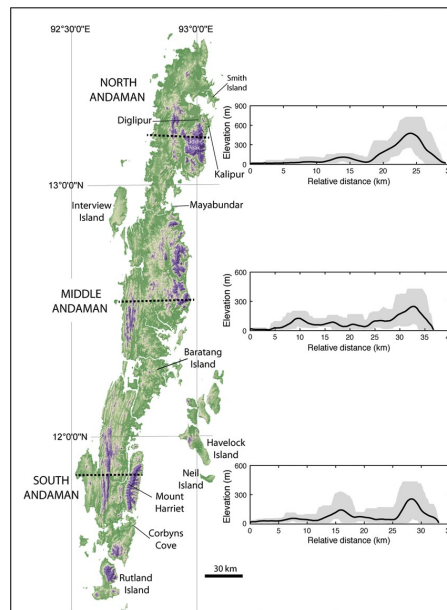


Fig. 2. Digital elevation model showing variations in the topography of the Andaman Islands. Source [16]

The islands do not have a very intense climate, with average temperatures within the range of 23 degrees Celsius to 31 degrees Celsius. The Islands receive rainfall through monsoons and a few tropical cyclones per year in the late summers. Coming to the discussion on the biodiversity of the islands, there exist more than 8300 species of fauna, out of which 1117 of the species are found only on these islands and even out of this, 64 of them are reptiles. Moreover, many of these species are critically endangered, such as the Shrews, Nicobar Owls etc. Out of the 270 species of birds, about one-third of them are endemic to the island. The Islands also house about 900 species of various insects which include about 225 species of butterflies. There are also more than 1000 species of Molluscs and Echinoderms in the sea and about 200 species of Corals. Moreover, a large variety of fish, such as saltwater fish and a few species of freshwater fish are found on the Islands as well as in the surroundings.

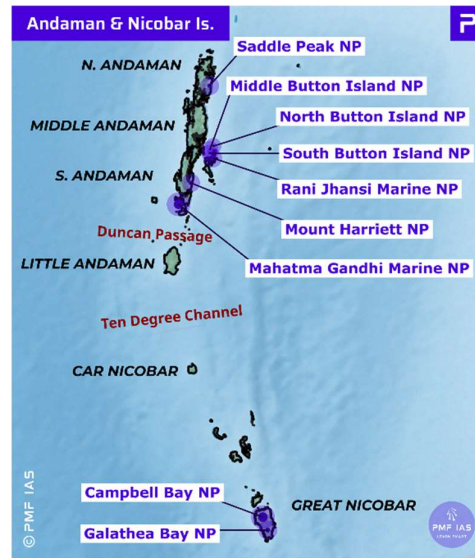


Fig. 3. Map displaying the National Parks in Andaman and Nicobar Islands. Source [17]

The flora of the island majorly consists of mangroves in marshes, coconut trees and tropical evergreen and deciduous forests. The north Andaman region has a larger density of wet evergreen forest along with creepers, whereas the middle Andaman region has wet deciduous forest, and the southern regions have epiphytic vegetation. About 86.2% of the total land area of the Islands is covered with forest and plants, with about 2200 species, out of which 200 are present only on this Island. Due to this rich biodiversity of the islands, there are 9 national parks, 96 Wildlife sanctuaries and one biosphere reserve in the Andaman and Nicobar Islands. Coming to the discussion on the demographics of the region as per the 2011 census, there are about 380000 people living on the island out of which about 600 are tribal people.

Apart from this, it is also important to look at the tectonic settings of the Andamans. The Andaman Islands are situated on the boundary lines closer to the Burma and Sumatra plates which are trying to slide over each other. This also forms a boundary of the Andaman Sea to the Bay of Bengal, leading to an almost disconnected body of water. Analysing the economy of the Andaman and Nicobar Islands, agriculture is the major source of occupation, with 50% engagement of the population; however, only 6% of the land has been used for agriculture. Other smaller sectors of the economy include fishing and registered small-scale industries working with handicrafts. The economy of the Islands is extensively dependent on the fishing industry which produced about 27000 tonnes of fish in the year 2017 from its exclusive economic zone of about

600000 km square area. Another major source of revenue is from tourism and according to a report from the NITI Ayog, the year 2016 experienced a total of 4 lakh visitors throughout the year.

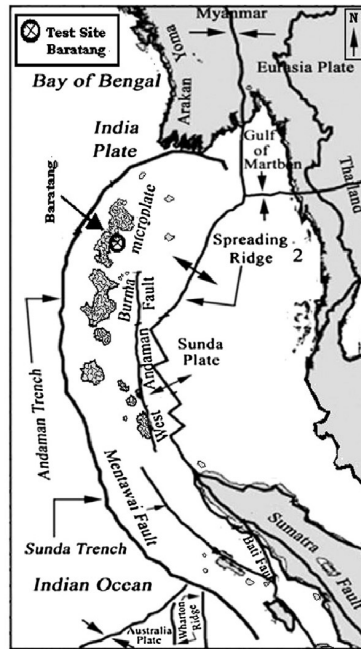


Fig. 4. Tectonic settings of the Andaman and Nicobar Islands (Modified after Mishra et al., 2007a). Source [18]

Despite this, the Andaman and Nicobar Islands have been facing a major impact of climate change over the course of time. A few of the major climatic events affecting the Andaman and Nicobar region are the rising sea levels, changing rainfall patterns, rising temperatures, growing construction and human activities, etc. which lead to major consequences on the Islands such as extreme weather events, drowning of the Islands, loss in property and life, loss in biodiversity, Coral bleaching, ecosystem degradation, decline in fisheries, etc. In the coming sections of the report, we will discuss all these factors and their effects in much greater detail, along with a simulation aimed to simulate the effect of sea level rise on lands of various types.

2 Major Factors Affecting Andaman

2.1 Sea Level Rise

Though sea level rise is a natural phenomenon, to a great extent it has been significantly accelerated by human activities. The sea level has risen by about 25 cm between 1901 and 2018 at an average rate of about 1 millimetre per year. However, this rate has been accelerated to 4.62 millimetres per year in the decade from 2013 to 2022, mainly attributed to anthropogenic effect.

Interestingly, the sea levels were about 25 cm lower than the present day in 1880, and there are certain projections of sea level fluctuation which even estimate that the sea levels may go as high as 2.4 m by 2100. As we had discussed in the introduction section, most of the Andaman Islands are quite low, with some rare areas reaching as high as 737 m.

Though there may be some natural factors, such as the Milankowitch cycle, which may contribute partly to this, the major reason is the anthropogenic effect attributed to three major effects: the first of them being Ocean heating. According to data, about 90% of heat generated as a result of Earth's energy imbalance finds its way into the ocean. This has raised the temperature of the ocean significantly. At the same time, it should have raised the atmospheric temperature by about 10 degrees Celsius. This increase in temperature of the ocean water has led to the expansion of the water and therefore, the rise in sea level. The other reason for this sea level rise is the significant ice loss in the Antarctica and Greenland regions.

According to some estimates, the earth's temperature has increased by about 1.5 degree Celsius with respect to the pre-industrial era, which has led to significant melting of ice in certain regions of Antarctica where it is estimated that about 12 Gt of ice from 2012 to 2016 has been lost. The Greenland ice sheet also has a significant contribution to increasing the sea level with some estimates mentioning that the contribution for the 2012 to 16 period from the Greenland ice was about 37% and the ice sheet has also lost about 3.3% of its volume in the duration from 2000 to 2019 which is an attribute to the fact that the average temperature in the Greenland region has increased by about 2.5°C with respect to the pre-industrial. Importantly, this level rise is not uniform all around the world, and this, in fact, is rising slightly faster in the Indo-Pacific regions such as Indonesia, Andamans, etc.

In the coming part of the report, I have done the simulation on the impact of sea level rise on different types of land in the Andaman and Nicobar Islands.

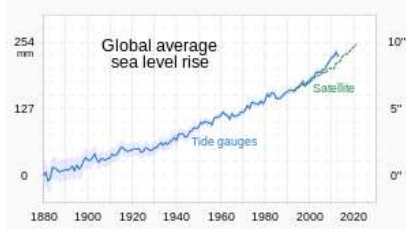


Fig. 5. The global average sea level rise curve, since 1880. Source [6]

2.2 Changing Rainfall Patterns

Due to their unique location between the Bay of Bengal and the Andaman Sea, the Andaman and Nicobar Islands have a very unique rainfall pattern. They mostly receive the rainfall through monsoons and cyclones in the late summer. Importantly, their location is also the reason for the vulnerability in their climate because it is projected that in the next 50 years, global temperatures are set to rise by about 2 to 3 degrees Celsius. Its impact will be tremendous on the Andaman and Nicobar Islands. This variation in the temperature has resulted in the creation of multiple acute cyclones outside the season of cyclones and the count of the cyclones is also significantly increasing along with their intensity. These intense cyclones are leading to anomalies in the rainfall of the region, which has many severe consequences, as we will discuss in the further sections of the report. According to the data at Port Blair city which is the capital of Andaman and Nicobar Islands, the mean annual rainfall is about 3079 mm per year. However, the variation in the rainfall is very significant at 15.8% in recent decades. This variation has led to huge unpredictability in the rainfall patterns, which also impacts various parts of the economy. Another trend from the region mentions that the total rainfall of the region is decreasing at an alarming rate of about 0.2 millimetres per year. Moreover, the average number of rainy days in the region used to be around 150 in the previous decade which has decreased to about 143. This has led to an alarming situation in the Islands, which has the potential to disrupt many sections of the ecosystem as well as the economy, as we will discuss in the coming part of the report.

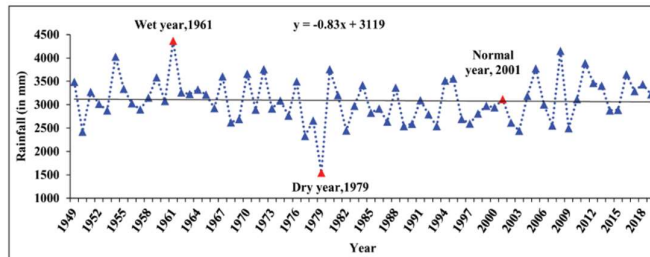


Fig. 6. Annual Rainfall variation in Andaman. Source [7]

2.3 Significant Rise in Temperature

There has been a significant rise in the temperature in some of the previous years compared to the pre-industrial era. This effect is known as global warming. There are a few natural factors working for this, but the contribution of the anthropogenic effects is widely accepted. Some of the factors, such as the rise in the usage of fossil fuels, deforestation on a huge scale and an increase in industrialisation, have led to this situation. The year 2023 was, in fact, the warmest year UN recorded, with a temperature of about 1.48 degrees Celsius, more than that with respect to the pre-industrial era. This global warming is not just a thing to worry about for now, but it may have multiple far-reaching consequences, with certain predictions predicting that the temperature of the earth may exceed about 3 to 5 degrees Celsius by 2100 with respect to the temperature in the pre-industrial era. One of the major contributors to this effect is the greenhouse gases and methane. According to reports, a large part of this carbon dioxide that is released into the atmosphere (about 22-30% of this) is absorbed by the oceans, which accounts for several gigatons of carbon dioxide consumed by the ocean per year. This has several impacts on the ocean, including the adverse effects of acidification on marine life. This leads us to the discussion on carbon emissions.

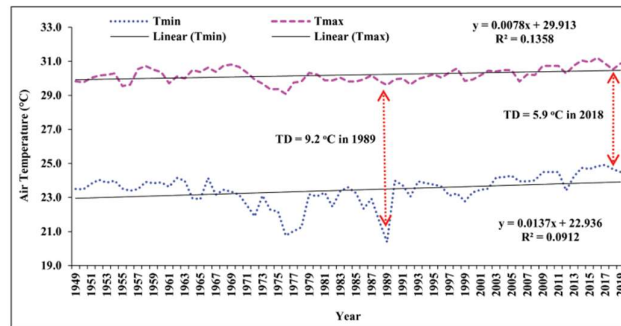


Fig. 7. Maximum and minimum temperature variation in Port Blair from 1949 to 2019. Source [7]

2.4 Extended Carbon Emissions

The carbon emissions, as well as the increased emission of other greenhouse gases, have significantly impacted the oceans. The data on carbon emissions suggests that the carbon content in the atmosphere has risen after the industrial revolution by anthropogenic contribution, and the rate has, in fact, accelerated in recent years. According to a few estimates, out of the Greenhouse gas emissions, about 75% of them were carbon-based, 18% were methane, 4% were nitrous oxides, and about 2% were fluorine-based gases. Moreover, the CO₂ that is emitted in the atmosphere lasts much longer than other gases such as methane. The increased emissions of carbon dioxide also have a huge tendency to disturb the biogeochemical cycle of carbon dioxide. Other sources of greenhouse gas emissions are other biological activities. The anthropogenic

effect of industrialisation has the greatest impact. There are many diverse effects of this extended carbon emission on the weather patterns as well as oceanic warming apart from the acidification of the ocean. These effects will be discussed further in the report.

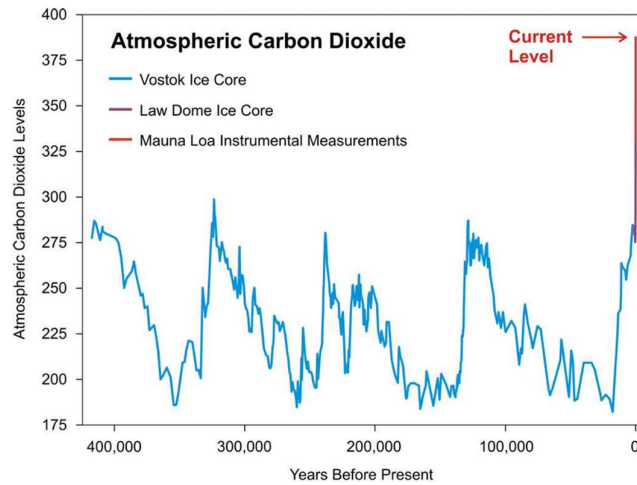


Fig. 8. Atmospheric carbon levels from 4,00,000 years before present. Source[10]

2.5 Growing Construction and Human Activity (Urbanisation)

According to official census data, the human population of the Andaman and Nicobar Islands has risen significantly. In the decades from 1951 to 1991, the population of the Islands almost increased by about 900%, whereas the National average was much less than 300%. Moreover, the Islands, which were home to about 30,000 people in the year 1951, are currently inhabited by around 4,00,000 people. Coming to the demographics of the islands, when the islands were first contacted, there were about 5000 native people whose count has reduced to about 44 great Andamanese, 15 Sentinelese and a few more people from other tribes. This indicates that there has been a decrease in the local population of native tribes as of 2016. Apart from societal reasons, we can also attribute this to climate change, which may have a tendency to threaten the habitat of these local tribes of the region. Due to the significant rise in the population of the Islands, there is also a huge demand for resources such as drinking water, and this has also resulted in a significant increase in the infrastructure of the island, which in turn has led to a threat to the biodiversity of the region. For example, the Great Nicobar Islands project, intended to develop infrastructure and cities has led to the clearing of 12-20 hectares of mangrove forests. This drastic destruction of biodiversity may lead to further worsening of the local climatic conditions of the region as the mangroves are the forests responsible for mitigating the effects of coastal soil erosion and harsh winds. The littering of beaches by plastics and other foreign products

also has a tendency to disrupt the ecosystem and, thereby, may also contribute to the local changes in the climate.



Fig. 9. Foreign litter on Great Nicobar Island. Source [19]

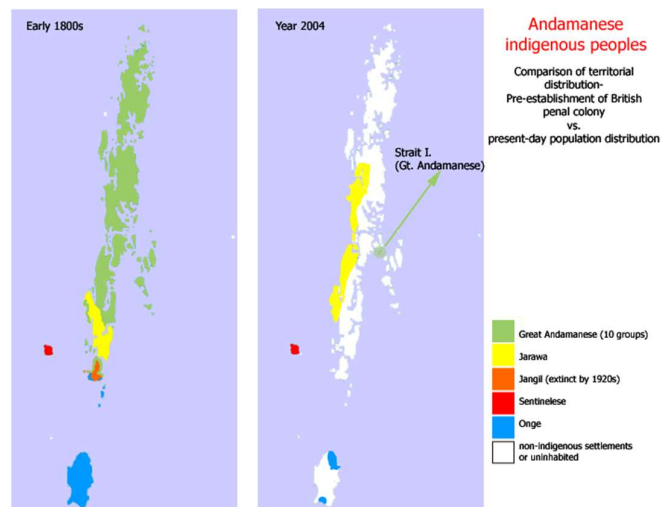


Fig. 10. Comparative distributions of Andamanese indigenous peoples, pre-18th century vs present-day. Source[20]

3 Major Impacts and Effects of These Events

3.1 Submerging of the Islands

The Andaman and Nicobar Islands are highly susceptible to the rising sea levels. The Eustatic sea level is on a steady rise. However, the relative sea level rise is also impacted by the tectonics and GIA. The cumulative effect of these results in the Relative Sea Level (RSL). The analysis in the simulations indicates that the net effect of these factors would result in a rise in RSL. Though the highest point of the island is at an elevation of 737 m, about 70% of the area of Car Nicobar Island and about 90% of the area of Great Nicobar Island is within 20 m level above the sea. Moreover, the elevation of some other islands that are currently inhabited indicates that over 15% of the total land area is within an elevation of 10 m above the current sea level. These are lands which are very susceptible to the rising sea level. Moreover, the region is in a very complex situation in terms of location due to the seismically active zones near the Sumatran and Burma plates. These islands are classified as the most severe seismic zone, zone 5 on the MSK intensity scale. There can be massive earthquakes that lead to significant changes in the topography of the island and may also lead to the subduction of Islands which significantly multiply the impact of sea level rise. According to studies, about 20% of the agricultural area is highly vulnerable to the rise in sea level. Apart from the gradual increase in sea level, there are much more destructive events that may be associated with the increase in sea level and surging of sea level within the coastal areas especially with the increasing heights of the tides leading to a significant increase in surges in the land, increase in the intensity of cyclonic storms in the region due to global warming, etc. The Andaman and Nicobar Islands are also susceptible to higher-rising Tsunamis due to their proximity to the seismically active zone. This has severe implications for the ecosystem as well as the livelihood of the people living in the region due to the coastal flooding. With the salt water of the ocean surging the coastal areas, there can be widespread destruction of infrastructure and human habitation, as well as the salination of the freshwater reservoirs. As discussed previously, about 20% of the agricultural land is highly susceptible to the increasing sea level which would lead to a significant loss of fertile lands for agriculture and grazing of the cattle. This would further lead to the clearing of the forest, thereby deepening the crisis of global warming. There are also estimated impacts of the rising sea level on the ecosystem of the Andaman Islands especially the flora of the region. Despite hosting a large diversity of flora, the Islands being susceptible to surge events may lead to the extinction of some of the species endemic to the island if enough measures are not taken in time. Moreover, there are even a few reports claiming that the Andaman Islands will be inhabitable by 2100 if enough care is not taken at present.



Fig. 11. Submerged parts of Campbell Bay, February 2006. Source[22]

3.2 Water Scarcity

Due to the small size, relative isolatedness, and topography of the Islands, there are very few freshwater reserves on the Islands, which are majorly rainfed. However, due to a significant change in the rainfall patterns of the region, as discussed previously, the sources of water have become extremely vulnerable to drought. Moreover, despite having large-scale development projects in the region, the focus on sustainable use of water is not much in these Islands. The groundwater is the main source of drinking water, but the water table is quite shallow, and the susceptibility from the mixing of saltwater with the freshwater resources of the region due to the rising in sea level is very high. Coastal flooding due to storm surges and tsunamis has further worsened the situation. This has a direct impact on the delicate ecosystem of the region, which depends on freshwater resources, as well as the economy of the region, which has a major dependency on agriculture.

3.3 Loss of Biodiversity and Ecosystem Degradation

Despite being a hotspot for biodiversity with a high degree of endemism, the Andaman Islands have suffered environmental degradation as a result of 150 years of resource exploitation without regard for conservation. Significant visible effects on the islands' ecology resulted from massive deforestation, infrastructure development, and the demands of an ongoing migrant and settler population. These effects included the extinction of numerous species, the introduction of invasive species, soil erosion, damage to the mangrove, coral reef, and littoral ecosystems, and the general upheaval

of a delicate ecological balance. Due to the significant rise in temperature of the region as well as increased human activities, the islands have experienced a significant loss of biodiversity. For example, one of the multiple infrastructure development projects, namely the great Nicobar Island development project, has led to a loss of 12 to 20 hectares of Mangrove vegetation in the area. It has also led to significant risk to the rare fauna of the region, such as leather Back Seat Turtles, Nicobar macaques, saltwater crocodiles, etc. Moreover, there are two new cities planned in this region which are expected to increase the population of the region by 350,000. Moreover, the intrusion of saltwater into the ecosystem of the area will lead to a significant loss of biodiversity, especially in the flora. The rising temperature of the sea, as well as acidification near the Andaman Islands, will further lead to a significant loss of diversity of fish in the area. The increase in fishing activities for hobbies and markets also poses a potent threat to biodiversity.

3.4 Coral Bleach

The Andaman and Nicobar Islands are extremely rich in the biodiversity of Marine ecosystems, with about 228 species of hard corals. There are also estimates that about 80% of the Global maximum number of the total Coral species are found in the Andaman Nicobar area. The shore waters are also rich in a huge variety of fish such as shellfish, film fish, etc. Major tourist attractions of these islands are also critically dependent on corals and fish. These are significant scientific resources for tracing the history of Oceans and the earth. However, the structure of these organisms is very delicate, and they are undergoing bleaching due to a significant increase in the temperature of the oceans, which will lead to a fundamental change in the coral reef habitat as well as an increase in the acidity of the oceans. According to a report, from April to July, the temperature rise was about 0.75 to 1.25 degrees higher than the rest of the months, which resulted in a mass bleaching of corals in the Andaman waters. Moreover, ocean acidification has also led to a significantly reduced Coral lifespan as well as a decrease in the Coral population of the island due to erosion by an increase in the intensity of cyclones. These lead to a significant vulnerability and threat to the Coral life of the region, which may also lead to an irreparable loss of biodiversity in the region.

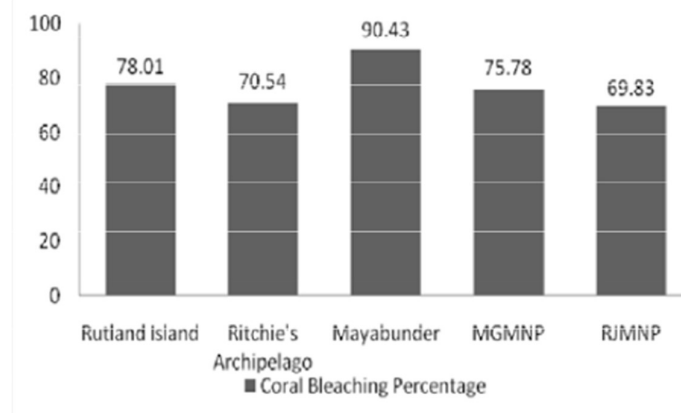


Fig. 12. Percentage of bleaching corals in study areas of Andaman Islands. Source [23]

3.5 Extreme Weather Events, Losses in Economy, Property, and Life

Though these islands contribute marginally to global climate change, they face most of the consequences of these global climate-changing events. There are continuous threats to the islands due to climate change and related disasters, such as cyclones, which are getting more and more frequent and violent, continuous rise in sea levels, tsunamis and earthquakes associated with the tsunami. One of the worst events in the history of Andaman was the 2004 Tsunamis, which brought huge devastation to the ecosystem as well as biodiversity and the economy of the islands, which had a major contribution from the fisheries, agriculture and tourism sectors. The tsunami led to a huge devastation of the fishing industry and also affected the habitats of the residents of the island. The extreme climatic events such as the cyclone and the changing rain patterns have also significantly impacted the agriculture sector of the region which is the major occupation of 50% of the population of the island. The degradation of the coral as well as the ecosystem, which is a major tourist attraction of the region, may also affect the tourism sector of the economy. The cyclone in the region has also led to significant destruction in the livelihoods of the people by destroying the crops, houses, roads, etc., as well as destroying other infrastructure of the city of Port Blair. Moreover, the heavy rainfall and the surge of seawater may also lead to flooding of low-lying areas of the Islands. This can lead to disruption in transportation across the Island by disrupting the road network and making the seas unsafe for travel. Frequent severe weather events can also lead to the suspension of seaport operations and airport services. Moreover, this can also result in the surging of saline water into the land-based freshwater resources such as tanks, lakes, ponds, etc., which may lead to a significant disruption in the agriculture sector of the region as the crops grown there, including paddy and cashew nut, require huge supplies of freshwater. This can lead to a significant disruption in the economy of the island. Erratic rainfalls in the region will further worsen the situation with longer summers, with higher temperatures and frequent droughts, which will lead to a severe summertime deficit of water in the region. Though drastic weather leads to

lower-scale destruction of the region as described above, we should also not forget the slowly occurring weather events and rise of the sea level, which threaten the Islands and the biodiversity of the region for a much longer duration.

4 Mitigation Strategies to Counter These Effects

Adaptability is the ability of an area of concern to accommodate or respond to changing conditions, which are also determined by the resources and technology available to the community (IPCC 2012). Capacity building is essential for individuals, communities and governments to work together regionally and internationally to make informed decisions about climate change adaptation and improve their long-term resilience. These include the protection of ecosystems affected by climate change and sea level rise and restoring ecosystems that have been damaged or destroyed by socio-economic development. In addition to the autonomous adaptation of natural coastal systems, one must also "plan" to recover and restore damaged mangroves and reef ecosystems.

Adaptation mechanisms aim to increase natural protection against sea levels and storms and provide resources for coastal communities. Some suitable adaptation strategies for island agriculture include agricultural diversification through efficient varieties of crops, modifications to site-specific integrated agricultural systems to diversify agricultural systems and adopt mitigation measures. The Integrated Agricultural System promises to improve the adaptive capacity of local communities by providing farmers with food, and income. Processing and value addition of agricultural products significantly reduces the vulnerability of island agriculture and stabilizes income. In order to save fresh water, we can adopt methods like rainwater harvesting for covered ponds, wells and dams. We should also try to understand and mitigate the scope of the salination of freshwater resources. Landscape methods can be used to address both land degradation and rainwater harvesting in coastal areas. We should also encourage local communities to actively participate in capacity-building and adaptation strategies. Development of awareness amongst the locals about the negative impact of climate change on their habitat and economy is the most important method. Involving the locals in sustainable construction activities as well as improvement in the "anti-flooding" infrastructure along with sufficient focus on the biodiversity wealth of the islands will be the key steps in the mitigation strategies. However, to make all this successful, we should have a meticulous collection of data and observations of changing trends in the region. We can combat these effects only by collaboration and coming together with the right use of science and technology.

5 Simulations

In order to get a better understanding of the climate change impact on the Andamans, I have performed a detailed analysis of the impact of sea level rise on various land types in Andaman. The digital elevation dataset was taken from SRTM. The data on the

various land use patterns was taken from the source mentioned in the references [1] section. The data was initially in .png format and .tiff format, which was extracted and transformed into a more convenient panda data frame. The corresponding code has been uploaded to the GitHub repository. The methodology takes into account various projections of sea level according to the models RCP 2.6, RCP 4.5 and RCP 8.5 and the projections are taken from 2050 till 2300. The analysis is done in the form of the percentage amount of land lost due to sea level rise till the year as given in the input. I also give the contour as well as surface plots of the remaining ‘unsubmerged’ areas of the region. The main assumptions taken here are that the GIA value in this region is about -0.1mm per year and the value of the rise of Andaman due to tectonics is about 1 mm per year, as discussed in class. These effects have not been considered for now and it is assumed that the sea level rise will be much larger than these values. This also means that there is a scope for a more accurate analysis of this region. Another important assumption is that the status of present-day land use is the same as 2300. That is, there is no deforestation done for agriculture and settlements. To use this application, first select the climate model which is needed to be used for the analysis. Then select the year from the slide bar. The simulation results are as displayed.

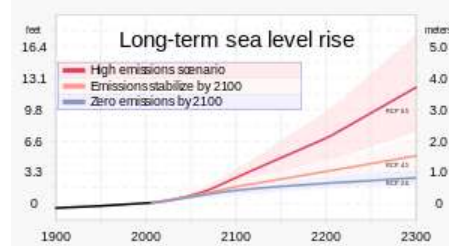


Fig. 13. Sea Level Projections (Till 2300) according to different climate models. Source [6].

For a better presentation of the simulation results, I have also created an online web application that would help in better interactive visualisation and understanding of the impact of sea level rise. A few major inferences are that the lands used for agriculture and settlements would be the most threatened by this increase in sea level. Mangrove forests may also face a similar threat from the rise in sea level. However, the forest lands and grass lands would be the least impacted ones. One more major observation is that the loss of land in the simulation with the RCP 8.5 model is much larger than the RCP 2.6 model, which implies that if we significantly reduce the emissions, we can control the damage and restrict the loss of land and property. However, the submergence of the settlement and agricultural land will further have a bad impact on the forest lands, which can be cleared for agriculture and settlements if sufficient actions are not taken in time. This would further worsen the situation, fuelling the current problem of loss of land by sea level rise.

The application can be accessed from the following QR code or from the Link given below. Due to the requirement of high computation, it may load slowly.



Fig. 14. QR Code for Accessing the Web Application of Simulation Results

Link for Accessing the Web Application of Simulation Results:

<https://ogc-climate-change-impact-on-andaman-modelling-xmzwv8nsg8xjuv.streamlit.app/>

Link for accessing the Code for the project: <https://github.com/JaidevSK/OGC-Climate-Change-Impact-on-Andaman-Modelling>

6 Conclusion

Worldwide, people are experiencing the unavoidable effects of climate change; however, different countries are more or less vulnerable to these effects depending on a number of variables. In the Andaman and Nicobar Islands, there is a noticeable shift in the seasonal distribution of rainfall and dry spells. Small islands are particularly susceptible to perceived climate change because they have lower adaptive capacities, mainly due to limited land and lower elevation. Rising sea levels may have an impact on coastal regions of these areas, especially the tourist destinations and larger settlements. Ecological habitats in a number of ways, such as eroding beaches, drowning low-lying areas, worsening coastal floods, and making freshwater aquifers more salinized. The main ways that this will show up are through habitat and biodiversity losses. Planned efforts will have the potential to improve communities' and ecosystem's resilience to climate change in addition to promoting natural adaptation.

Importantly, in order to strengthen adaptation measures, there is an urgent need to advance scientific knowledge and the capacity to forecast the effects of rising sea levels. Lessening the effects of sea-level rise on these islands' economy and environment may

need to include sea-level rise into coastal planning and developing tools to help decision-making for additional adaptive action.

7 Dataset Credits and Reference:

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